

Retrieval Augmented Generation

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Model Memory vs Retrieval

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Speech and Audio Systems

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Building a grounded answer. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Building a grounded answer. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Sources: A='Library opens 9'; B='Cafe opens 8'; C='Library closes 5'. A question asks when the library opens. The retriever returns only B and C. What should the generator do?

6. Solve the given case

Note A says library opens at 9. Note B says returns go in the blue box.

Answer opening time with evidence.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A retrieved passage says the club meets Tuesday; it says nothing about cost. The assistant answers Tuesday, free entry. Which part is grounded and what should replace the unsupported part?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 18 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Sources: A='Library opens 9'; B='Cafe opens 8'; C='Library closes 5'. A question asks when the library opens. The retriever returns only B and C. What should the generator do?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Chunking and document structure — W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

18. Chunking and document structure — W2 Trace the worked case

A notice says capacity 20, followed by during maintenance capacity 8. Separate chunks contain each sentence but retrieval returns only the first. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

19. Chunking and document structure — W3 Audit the limit

A report omits this qualification: Smaller chunks can improve selectivity while losing necessary qualifiers; chunk size alone is not retrieval quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

20. Chunking and document structure — W4 Apply a new case independently

A rule section says rooms close at 17:00; an exception paragraph says Friday closes at 16:00. The question is Friday closing. Design a complete chunk reference.

My reasoning and evidence

21. Chunking and document structure — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

22. Lexical and vector retrieval — W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

23. Lexical and vector retrieval — W2 Trace the worked case

Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector query $[1, 0]$, $D1=[.4, .9]$, $D2=[.8, .1]$. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

24. Lexical and vector retrieval — W3 Audit the limit

A report omits this qualification: Similarity and relevance are different quantities; exact terminology, metadata and permissions may be necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

25. Lexical and vector retrieval — W4 Apply a new case independently

Query terms blue battery. E1 text blue battery guide, vector $[.2, .8]$; E2 text blue ribbon, vector $[.9, .1]$. Query vector $[1, 0]$. Compute both rankings.

My reasoning and evidence

26. Lexical and vector retrieval — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

27. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

28. Pictorial case practice: Retrieval Augmented Generation

Turn a short science source into an answer with a visible claim-to-source map.

Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Retrieval Augmented Generation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read source

S4 supports the material identity and the solid-to-liquid change.



2. Draft two claims

Draft: ice is solid water; warming can turn it into liquid water.



3. Check each claim

Both claims match S4. A claim that melting takes exactly two minutes would not match.

CHANGE ONE THING

The question asks exactly how long this ice takes to melt.

Explain the changed result using the matching lesson.

Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

29. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

